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Attention-assisted hybrid CNN-BILSTM-BiGRU model with SMOTE–Tomek method to detect cardiac arrhythmia based on 12-lead electrocardiogram signals

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Abstract

Objectives: Cardiac arrhythmia is one of the most severe cardiovascular diseases that can be fatal. Therefore, its early detection is critical. However, detecting types of arrhythmia by physicians based on visual identification is time-consuming and subjective. Deep learning can develop effective approaches to classify arrhythmias accurately and quickly. This study proposed a deep learning approach developed based on a Chapman–Shaoxing electrocardiogram (ECG) dataset signal to detect seven types of arrhythmias.

Method: Our DNN model is a hybrid CNN-BILSTM-BiGRU algorithm assisted by a multi-head self-attention mechanism regarding the challenging problem of classifying various arrhythmias of ECG signals. Additionally, the synthetic minority oversampling technique (SMOTE)–Tomek technique was utilized to address the data imbalance problem to detect and classify cardiac arrhythmias.

Result: The proposed model, trained with a single lead, was tested using a dataset containing 10,466 participants. The performance of the algorithm was evaluated using a random split validation approach. The proposed algorithm achieved an accuracy of 98.57% by lead II and 98.34% by lead aVF for the classification of arrhythmias.

Conclusion: We conducted an analysis of single-lead ECG signals to evaluate the effectiveness of our proposed hybrid model in diagnosing and classifying different types of arrhythmias. We trained separate classification models using each individual signal lead. Additionally, we implemented the SMOTE–Tomek technique along with cross-entropy loss as a cost function to address the class imbalance problem. Furthermore, we utilized a multi-headed self-attention mechanism to adjust the network structure and classify the seven arrhythmia classes. Our model achieved high accuracy and demonstrated good generalization ability in detecting ECG arrhythmias. However, further testing of the model with diverse datasets is crucial to validate its performance.

Keywords

Arrhythmia detection, deep learning, electrocardiogram, CNN, BILSTM, BIGRU, SMOTE–Tomek

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Introduction

Cardiovascular diseases (CVDs), which impose a significant burden on individuals and societies, are among the leading causes of mortality and morbidity worldwide.[1] Among the various CVDs, arrhythmia is one of the most severe and can be fatal.[2] Arrhythmia refers to irregularities in the rate or rhythm, such as too fast (tachycardia), too slow (bradycardia) or irregular heartbeat patterns.[3] Cardiac

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arrhythmias vary from benign rhythms to potentially life-threatening rhythm types.[4] Therefore, early detection of arrhythmia is critical for clinical practitioners to save human lives.[5]

The electrocardiogram (ECG) is a well-known visual time-series diagnostic method for arrhythmias that records the electrical activity generated by each cardiac cycle in real-time and is now widely used in heart rate detection.[6,7] The standard clinical format of the ECG is a 12-lead ECG, including bipolar leads I, II and III; enhanced unipolar leads aVR, aVL and aVF; and unipolar leads V1 through V6.[8] The ECG signals are captured from a different angle activity in both horizontal and vertical planes when the electrodes of an electrocardiograph device are attached to the patient's skin. Ten electrodes are needed to capture 12 leads (signals) and provide an accurate representation of the heart's rhythm.[8,9]

Correctly detecting the type of arrhythmia is crucial for physicians before administering treatment because it not only helps save a patient's life but also alleviates sequelae, thereby reducing the burden and cost of healthcare.[10] The current standard method for detecting arrhythmia types is visual identification, which can lead to physicians' subjective biases. Due to the large morphological variances, it is not simple to manually detect ECGs.[11] Additionally, the diagnosis and interpretation of a wide variety of CVDs by ECG mostly rely on physicians' knowledge and expertise.[12] However, physicians' diagnosis of cardiac disorders is always associated with levels of imprecision.[4] Therefore, specialists have tried to develop effective approaches to the early diagnosis of cardiac disorders through strong deep learning. Various researchers have attempted to develop effective approaches to accurately and quickly classify arrhythmias using strong deep learning methodologies.[9]

Deep learning techniques overcome the constraints imposed by traditional diagnosis methods, enhancing performance and generalization by decreasing pre-processing and feature extraction.[13] Deep learning (DL) is a series of representation layers[14] involving the automated process of detection and extraction of important features and self-learning through training processes, which makes it a very accurate approach.[15,16] According to a study, arrhythmia classification by deep learning techniques can achieve higher efficiency and accuracy than expert manual detection.[10,17] Among deep learning techniques, especially convolutional neural networks (CNN), recurrent neural networks (RNN), which primarily contain long short-term memory (LSTM), and gated recurrent unit (GRU) networks have been particularly successful in ECG signal processing and were significantly improved.[18]

CNNs are among the most widely used DL architectures made to learn the spatial hierarchy of data adaptively by extracting and remembering high-level and low-level patterns to predict the output.[19] One-dimensional convolutional neural networks (1D CNN) were developed to use one-dimensional signals, such as ECG records.[20,21] Another deep learning algorithm for ECG analysis is LSTM networks that can learn from sequential data, known as sequence learning.[5] LSTM is an enhancement of the RNN memory performance. Many studies have been conducted on the applicability of LSTM to time-series problems.[22,23]

Bidirectional LSTM (BiLSTM) is an improvement of LSTM, and the models offer better predictions than LSTMs.[24] Most researchers combined CNN with BiLSTM and achieved satisfactory results in classifying arrhythmias from ECG signals.[10] The GRU network is another type of recurrent neural network, which is similar to an LSTM network but has fewer parameters and converges faster than LSTM. GRU adaptively remembers and forgets its state based on the input signal to the unit.[25] A bidirectional gated recurrent unit (BiGRU) consists of two GRU layers and can process the data in both directions.[25,26] In general, different DL techniques based on varied datasets focusing on the automatic detection of ECG arrhythmia classification have been proposed in the past few years.[27]

There is usually some degree of imbalance between different classes in any ECG dataset, which is an additional challenge to accurately classify ECG beats.[28] Dataset imbalance affects the model's training. Generally, classes with more samples can dominate the training and cause the classifier to favor classes containing many labeled samples.[29] This leads to low training efficiency because normal ECG beats occupying a large proportion of the dataset are prone to negative effects and model degeneration.[30] To address this problem, the model design must use a loss function in which the cost of misclassifying the minority class is much higher than that of misclassifying the majority class, like the focal loss (FL) function.[31] It is still challenging to deal with an imbalanced dataset using deep learning for classification of multiple types of cardiac arrhythmias, which is highly dependent on data quantity and quality. Furthermore, the high noise and complexity of ECG place a substantial strain on computational resources; therefore, more robust models with fewer parameters are greatly desired.[10]

In this study, we designed a deep learning model to classify arrhythmia into seven classes: atrial flutter (AF), atrial fibrillation (AFIB), sinus irregularity (SI), sinus bradycardia (SB), sinus rhythm (SR), sinus tachycardia (ST) and supraventricular tachycardia (SVT) by the 12-lead ECG arrhythmia database collected by Shaoxing People's Hospital (Shaoxing Hospital Zhejiang University School of Medicine) and Chapman University on severely imbalanced data.[32]

Our proposed model consists of designing a hybrid model with a fusion of convolutional blocks, a bidirectional RNN (BILSTM-BiGRU) block and an attention mechanism. The second main contribution of this study was providing a solution for the imbalanced data problem by the hybrid synthetic minority oversampling technique (SMOTE)–Tomek dataset balancing algorithm and FL function to resolve the data

imbalance problem for detecting seven types of arrhythmias from ECG signals.

Related Literature

Different DL techniques based on varied datasets focusing on the automatic detection of ECG arrhythmia classification have been proposed in the past few years.[27] Nevertheless, the analysis of the findings from each deep learning technique depends on various factors, such as the hardware platform, the model's architecture and compiler optimization, all of which can directly affect the model's training.[33] Deep learning models can extract more distinguishing features automatically and without the need for hand-craft feature extraction to facilitate the detection of the arrhythmia pattern in ECG signals.[21,27] In recent years, there has been extensive research using deep learning methods for ECG classification and arrhythmia detection based on ECG datasets, such as the MIT-BIH, Chapman ECG, PTB-XL and several other datasets.[34] In this section, we review the most important deep learning techniques presented for detecting arrhythmia based on ECG signals. Various studies introduced CNN-based deep methods that use 1D filters. Here, several CNN-based methods are explained.

Romdhane et al. designed a CNN classifier using the FL function. The model was trained and evaluated with the MIT-BIH and INCART datasets to identify five arrhythmia categories. The classifier used the dropout and batch normalization mechanism in a coupled-convolution layer structure, obtaining an overall accuracy of 98.41%.[29] Wu et al. proposed a robust and efficient one-dimensional 12-layer convolution neural network (CNN) network structure to classify the five sub-classes of cardiac arrhythmia. The architecture of the CNN network included eight alternating convolutions and average-pooling layers. They were followed by a dropout layer and two fully connected layers in the MIT-BIH arrhythmia database (MITDB), and an overall performance accuracy of 97.2% was achieved.[35] Ahmed et al. developed a novel model consisting of three convolution blocks for ECG arrhythmia classification. Each block contains two 1D-CNN layers, a max-pooling layer, a dropout layer and a batch normalization layer tested on the MITDB. The model achieved overall performance accuracy of 1.00 and 0.99 in the training and testing datasets, respectively.[36] Gao et al. designed a classifier based on effective LSTM recurrence network model with FL to achieve imbalanced ECG signal classification. The model attained an overall accuracy of 99.26% for the categorization of eight beat types in the MITDB, demonstrating outstanding performance.[30] Yildirim et al. proposed a model which included both representation learning and sequence learning tasks. Convolutional layers and sub-sampling layers were used in the representation learning phase. The sequence learning part involved a LSTM unit after the representation of learning layers. They performed two class scenarios, including reduced rhythms (seven rhythm types) and merged rhythms (four rhythm types) according to the records from the Chapman ECG database. The dataset was randomly divided into training, validation and test sets. The model consisting of six convolution layers bearing one-dimensional filters, four max-pooling layers, one LSTM and two fully connected layers categorically used the cross-entropy loss function and Adam optimizer for adjustment. The trained DNN model achieved an accuracy of 92.24% and 96.13% for the reduced and merged rhythm classes, respectively.[5] Rai et al. proposed two distinct deep learning models: CNN and a hybrid model, CNN-LSTM, to detect six types of cardiac arrhythmias. Two standard datasets MIT-BIH arrhythmias database (MITDB) and PTB diagnostic database (PTBDB) were combined, and 23,998 ECG beats were extracted for the model validation. Imbalanced datasets were addressed by using a hybrid data resampling technique called SMOTE and Tomek. The highest overall accuracy was 99.02% on the SMOTE–Tomek sampled dataset by the ensemble technique.[37]

Most researchers combined CNN with BILSTM and achieved satisfactory results in classifying arrhythmias from ECG signals. Hassan et al. proposed a novel deep learning model that combined CNN and bi-directional LSTM to classify five classes: non-ectopic (N), supraventricular ectopic (S), ventricular ectopic (V), fusion (F) and unknown (Q) beats. The proposed model was trained, validated and tested using MIT-BIH and St Petersburg data sets separately. This model achieved training, validation and testing accuracies of 100%, 98% and 98%, respectively.[6] Islam et al. proposed the CNN and bidirectional RNN unit (BiGRU–BiLSTM) architecture to generate fusion features. The signals were classified by the fully connected layer and rectified linear unit (ReLU) activation function. The model was measured by the categorical cross entropy (CCE) loss. The experimental findings to detect five classes showed that, for the MIT-BIH-provided ECG data to identify arrhythmias, the proposed BRDC model outperforms the existing models with an accuracy of 99.90%.[38] Islam et al. developed a novel hybrid hierarchical attention-based bidirectional recurrent neural network with dilated CNN (HARDC) to detect five types of arrhythmias from ECG signals using 48 half-hour ECG records of patients investigated at the BIH Arrhythmia. The method achieved an accuracy of 99.60% during training, 99.40% during validation and 99.01% during testing.[39]

A study by Rai et al. proposed an automated system for detecting myocardial infarction (MI) using ECG signals. Their approach involved a hybrid model combining CNN and LSTM, along with an ensemble technique. The model was trained and evaluated using the PTBDB and the MITDB. The CNN achieved an impressive classification accuracy of 99.82%, while the hybrid CNN-LSTM and

ensemble techniques achieved even higher accuracies of 99.88% and 99.89%, respectively.[40] In another study by Rai et al., a hybrid CNN-LSTM deep learning model utilizing the PTBDB was proposed for the prognosis of MI. The model leveraged ECG beat time intervals and their gradient values as input features. To address the class imbalance, the researchers employed the SMOTE–Tomek link data sampling technique. The CNN-LSTM model achieved a remarkable accuracy of 99.8%.[41] Furthermore, in a study conducted by Rai et al., a hybrid CNN-LSTM deep learning model was proposed for the prediction of cardiac arrhythmias. The model was trained and evaluated using the PTBDB and the MITDB. The model achieved a high accuracy of 99.7%.[42] Sadeghi et al. in 2023 introduced CNNs with SE-Residual blocks and a self-attention mechanism as important components for a more accurate diagnosis of left bundle branch block. This model was evaluated on the PhysioNet Challenge 2020 database, consisting of 10,344 samples of 12-lead ECGs, using a 10-fold cross-validation approach. The model achieved an accuracy of 91.98%, a specificity of 28.99%, a precision of 73.09%, a sensitivity of 82.83%, an F1 score of 77.99% and an area under the curve (AUC) of 0.991.[43]

Methods

Study duration and place

The study was conducted at the Iran University of Medical Sciences (IUMS) located in Tehran, the capital of I.R. Iran. The research took place from August 2019 to May 2022.

Nature of the study

This was a developmental study, carried out quantitatively. The study proposes a deep learning approach developed based on a Chapman–Shaoxing ECG dataset signal to detect seven types of arrhythmias. The basic steps of our proposed methodology are presented using the flowchart in Figure 1.

Data processing

Dataset. This study used the Chapman–Shaoxing ECG dataset. This dataset contains 12-lead ECG signals collected from 10,646 people for 10 s with a frequency of higher than 500 Hz sampling rate under the supervision of Shaoxing People's Hospital (Shaoxing Hospital Zhejiang University School of Medicine) and Chapman University. Each ECG sample in this data set contains 5000 rows and 12 columns, labeled by experts in 11 types of heart rhythms. Each column corresponds to a specific lead, including Lead I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5 and V6. With a sampling rate of higher than 500 Hz, each of the 500 rows in a Microsoft Excel column represents the data recorded for one second of the patient's heart function.

The interesting point about this dataset is that the signals are recorded from patients over several days and during different sessions.[32,44] Table 1 demonstrates numerical information about the ECG dataset.

The Chapman ECG dataset was a valuable resource for our research on arrhythmia detection, primarily due to its unique characteristics. With 12-lead ECG samples and a sampling frequency of 500 Hz, the dataset is well-suited to our research objectives. By utilizing this dataset, we can effectively train algorithms and models for accurate arrhythmia detection using real and reliable data. The selection of the Chapman ECG dataset aligns seamlessly with the timeline of our research proposal, which was approved in 2019, as it was the most suitable option among the available datasets at that time. Moreover, the public availability of the Chapman ECG dataset ensures the reproducibility of our research, facilitating further validation and comparison of the findings. Considering these factors, we are confident that our decision to employ the Chapman ECG dataset as the primary dataset for our research is entirely justified.

Data denoising method. This study implemented the sequential denoising approach to the raw ECG data, according to the approach proposed by Zheng.[32] Initially, a Butterworth low-pass filter was used to remove the signals with a frequency of above 50 Hz.[45] Then, the local polynomial regression smoother (LOESS) was utilized to remove the effects of baseline wandering.[46] Finally, non-local means (NLM) were utilized for residual noise reduction.[47,48] These processes were sequentially used to process the raw ECG samples. Figure 2 demonstrates an instance of an ECG signal. Figure 2(a) presents the raw sample of this ECG, Figure 2(b) displays the noise reduction performance after the Butterworth low-pass filter, Figure 2(c) displays the effectiveness of the LOESS smoother and Figure 2(d) indicates this ECG sample after NLM at the end of noise removal preprocessing operations. Generally, the preprocessing operations help normalize signals and reduce complexities. After noise removal preprocessing operations, ECG signals with frequencies above 50 Hz and the baseline wandering effect were removed. Since some ECG samples' lead values were missing and some contained only zeros, these samples were removed, and a total of 10,618 samples from this dataset were used.

Data balancing. The Chapman–Shaoxing dataset used is an imbalanced ECG dataset.[32] Imbalanced datasets can result in biased and poor performance of the model for minority classes because the model's training is heavily inclined toward the majority class through the backpropagation of the loss function.[10] To prevent the loss function from being disproportionately influenced by the larger sample category in the imbalanced dataset, we implemented the SMOTE–Tomek algorithm. First, SMOTE, which is an oversampling method, increases minority class samples so that the minority class samples are expanded. This

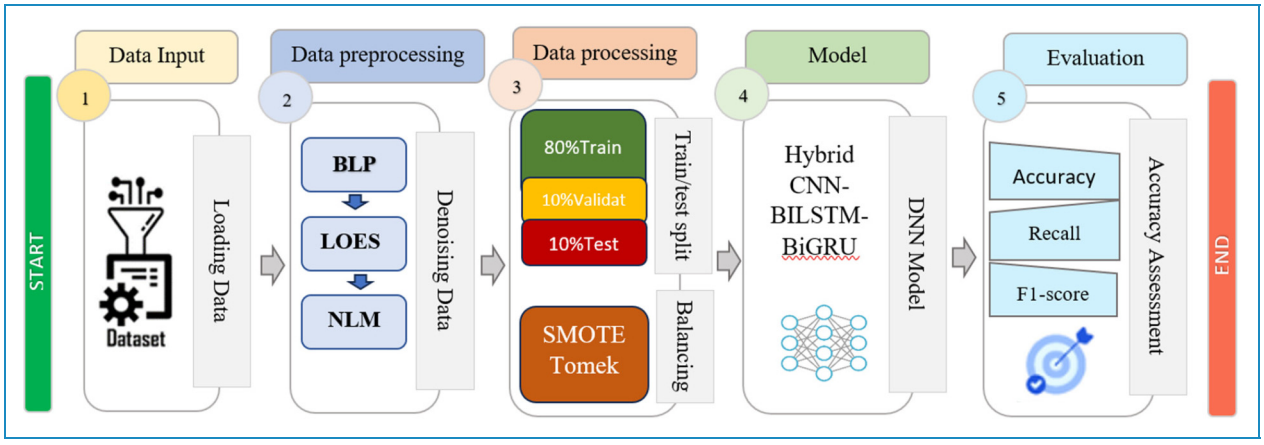


Figure 1. The methodology followed in this study.

Table 1. Numerical information about the ECG dataset.

Acronym	Full Name	Frequency N (%)
SB	Sinus bradycardia	3889 (36.53)
SR	Sinus rhythm	1826 (17.15
AFIB	Atrial fibrillation	1780 (16.72)
ST	Sinus tachycardia	1568 (14.73)
AF	Atrial flutter	445 (4.18)
SI	Sinus irregularity	399 (3.75)
SVT	Supraventricular tachycardia	587 (5.51)
AT	Atrial tachycardia	121 (1.14)
AVNRT	Atrioventricular node reentrant tachycardia	16(0.15)
AVRT	Atrioventricular reentrant tachycardia	8 (0.07)
SAAWR	Sinus atrium to atrial wandering rhythm	7 (0.07)
Total		**10,646 (100)**

Bold value represents the best performance.

synthesizes new plausible examples in the minority class, but the new samples generated by SMOTE also have some limitations.[49,50] The newly generated samples have distinct boundary limitations. The Tomek Links algorithm was combined to achieve data balance[50] to avoid this problem caused by oversampling. The Tomek Links algorithm is an under-sampling algorithm for identifying pairs of nearest neighbors in a dataset that have different classes. SMOTE–Tomek is a combined over- and under-sampling using SMOTE and Tomek links.[51] SMOTE–Tomek has been shown to be effective in mitigating the effects of class imbalance.[37] After conducting a review, it was found that none of the previous studies utilized the SMOTE–Tomek technique on the Chapman ECG dataset. However, due to the favorable results reported in using this technique to reduce data imbalance in other studies, we decided to adopt this method. The proven performance of this technique in similar scenarios supports its selection as a suitable approach to address the challenge of class imbalance in the Chapman ECG dataset.

Model architecture

In the proposed approach, we designed a novel hybrid model for automatically detecting seven types of heart arrhythmias, which helps improve overall performance. The overall structure of the proposed deep learning model is implemented by combining two neural networks, CNN and bidirectional RNN (BILSTM-BiGRU). The choice to integrate these two techniques was made with the understanding that their combination provides better outcomes.[52] The model parameters' settings were adjusted to provide the optimal result. In addition, issues such as which layers should be used and which parts of the model should be placed were time-consuming and difficult processes solved by trying many variations.

Convolutional neural network blocks. One-dimensional convolutional neural networks (1D CNN) are a variation of two-dimensional CNNs (2D CNNs) developed for using one-dimensional signals such as ECG records.[20,21] CNN may be capable of retrieving valuable data whenever the input signal is noisy. The network structure is constructed

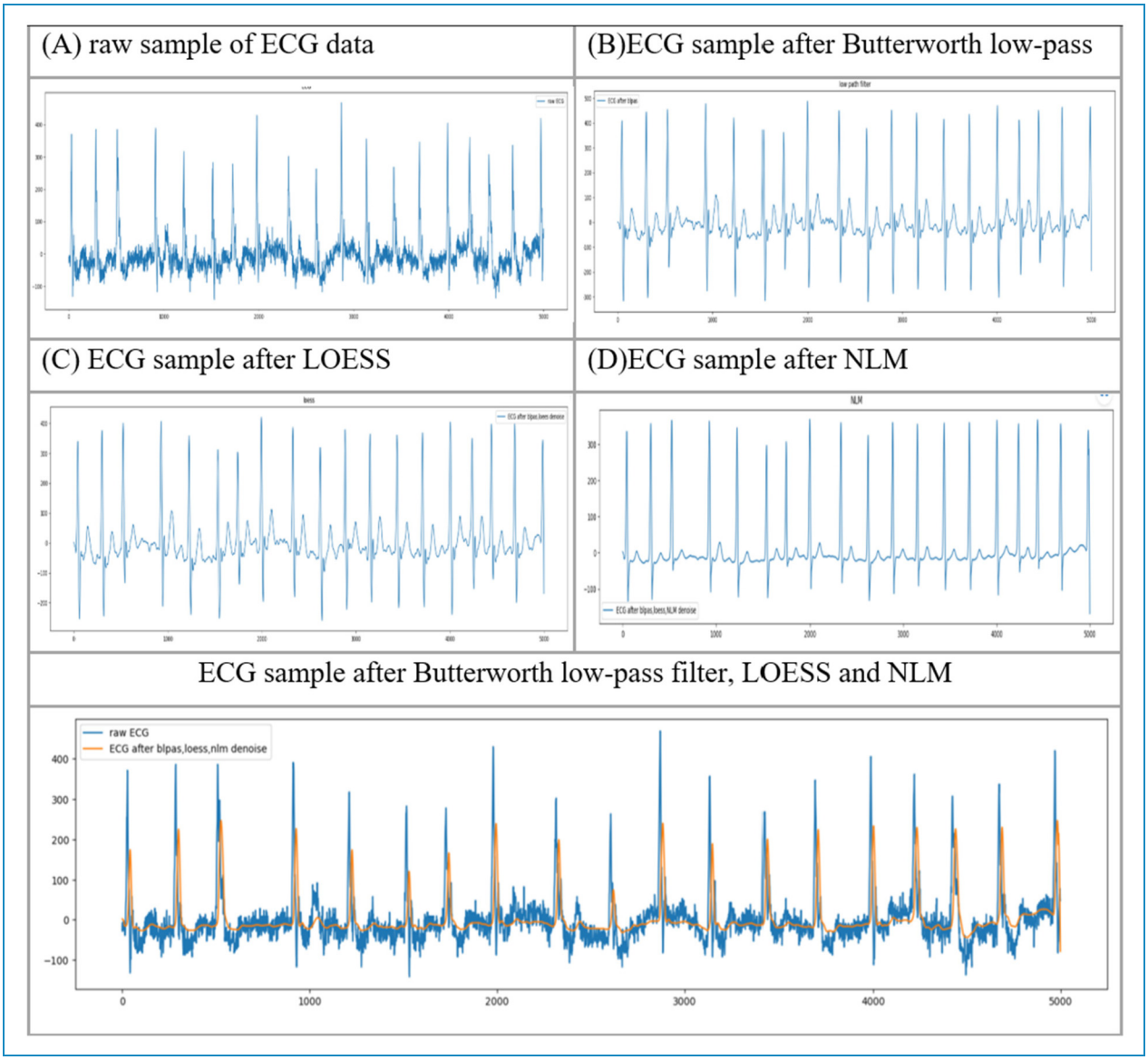


Figure 2. An instance of ECG signals.

layer by layer to reflect these performance characteristics. As the amount of network layers increases, features are learned and expressed in a more abstract and concise manner.[6,53] The raw data of ECG signals of seven heartbeat types after preprocessing in the training dataset are input into the three convolutional blocks, each of them consisting of two Conv1D layers.

Conv1D can learn to distinguish hierarchical features from raw inputs.[5] To determine the correct parameters, such as the number of filters, kernel size and strides, we used the experience of Yildirim's study.[5] A leaky rectified linear unit (Leaky ReLU) activation function was used in each Conv1D layer. Activation functions are crucial to increasing the expressiveness of neural networks and enhancing the approximation capability between the network's different layers.[36,52] To boost the effectiveness of feature extraction from the raw ECG signals, one can add a 1D max-pooling layer of size 2 after each convolutional block to reduce the computational cost. The pooling layer is a significant step and is commonly the next layer of convolution, which is also called the subsampling layer. The max pooling operation divides the input data into rectangular regions and retrieves the maximum value from each subregion, thereby reducing the number of neurons and the dimensionality of the convolution layer output data. This reduction in network complexity helps prevent overfitting and enhance the network's robustness.[35,54] A batch normalization layer was used to normalize the data with a dropout layer rate of 40% placed at the end of each block to avoid the over-fitting problem. The

dropout layer is used when training the CNNs to temporarily disable some neurons from the network with a certain probability. This reduces the interdependence between neuron nodes, mitigates overfitting and improves the network's generalization capability.[35]

Adaptive max pooling. Adaptive pooling reduces the amount of data needed in tuning the downstream portion of CNNs. This technique is utilized to perform multiscale summarization over convolutional feature maps while capturing the required behavior of the feature map itself, decrease the convergence time, and enhance generalization performance.[55]

Bidirectional RNN (BILSTM-BiGRU). After passing through the 1D-CNN layers, the output characteristics are fed into the bidirectional RNN (BILSTM-BiGRU) block consisting of a BiLSTM and a BiGRU to analyze sequential ECG. It is proven that bidirectional RNNs can learn sequential features from ECG data in both forward and backward directions.[56] BiLSTM is an improvement of LSTM that enables additional training by traversing the input data twice, once in the input data (forward) and once in the reverse input data (backward). BiLSTM models offer better predictions than LSTMs, although LSTMs reach equilibrium faster than BiLSTMs.[24] The application of convolutional operations to the input matrix before applying the BiLSTM model is a promising approach. Additional geographical and temporal data properties may be incorporated, thereby increasing the accuracy of predictions.[57] The GRU (gated recurrent unit) network is another type of recurrent neural network that resembles an LSTM network but has fewer parameters and converges faster. GRU remembers and forgets its state adaptively based on the input signal to the unit. In terms of performance, the difference between the two is often small and inconclusive, depending on the task and the dataset.[58] A BGRU consists of two GRU layers and can process data in both directions: one processing the input sequence from left to right and the other from right to left, with two separate hidden layers. The outputs of the two GRU layers are concatenated at each time step, forming a bidirectional representation of the input sequence. BIGRU can capture both the past and future contexts of a sequence and can improve the performance of tasks that require long-term dependencies or complex structures.[25,26] Each layer has 128 neuron units and a dropout of 40%, which can alleviate the overfitting concern.

Multi-head self-attention mechanism. RNN compresses all information over time, which can weaken the time difference between input features and affect prediction accuracy. The attention mechanism is used to focus on specific parts of the input sequence based on their importance.[59] The output of the BILSTM-BiGRU layer is transmitted into the self-attention layer. The attention layer is primarily used to focus on a particular network layer. The multi-head self-attention mechanism allows the model to learn relevant information in different subspaces of representation, and it becomes possible to capture a broader range of discriminative features. The self-attention mechanism can process the input data in parallel, reducing the complexity of calculations.[60,61]

Fully connected layer. Finally, the predicted model consists of a flattening layer and a dropout layer, whose extracted features are used as the input for a dense layer with a Relu activation function and 64 neurons. The dropout layer is set to discard 40% of the input features to prevent overfitting. The output of the dense layer is then fed to a SoftMax layer, which predicts the probability of the ECG heartbeat class.

Model compile. This study utilized FL cross entropy with alpha = 0.25 and gamma = 2 to calculate the loss in the model's training and validation. FL is a modification of cross-entropy (CE) loss where the scaling factor decays to zero as the confidence of the classification increases; as a result, the loss assigned to well-classified examples is down-weighted.[30,62] The FL approach reduces the impact of imbalanced heartbeat classes by focusing the loss on minority classes. This technique has been shown to improve the performance of deep learning models on imbalanced datasets.[29]

Model checkpoint. The model checkpoint option, provided by Keras, automatically saves the weights of the best suitable model in terms of a specified metric.[7,63] We choose to monitor the loss value evaluated on the validation data to keep the model's weights with the monitor. The schematic illustration of the proposed model to classify the ECG signals of this study is illustrated in Figure 3. This diagram depicts in detail the fundamentals of each layer, from the source term to the accurate prediction layer. The arrhythmia goal category is predicted at the end of the process. The structure of the constructed CNN-BILSTM-BiGRU model is depicted in Table 2.

Results

A comprehensive experimental analysis is described in this section.

Experimental setup

We have designed a hybrid architecture model to effectively classify imbalanced arrhythmia signals. The experimental analysis was conducted on the Chapman–Shaoxing ECG dataset. We excluded certain classes from the initially published dataset due to an insufficient number of cases. Specifically, the classes atrial tachycardia (121 samples), atrioventricular node reentrant tachycardia (16 samples), atrioventricular reentrant tachycardia (8 samples) and sinus atrium to atrial wandering rhythm (7 samples) were removed from the

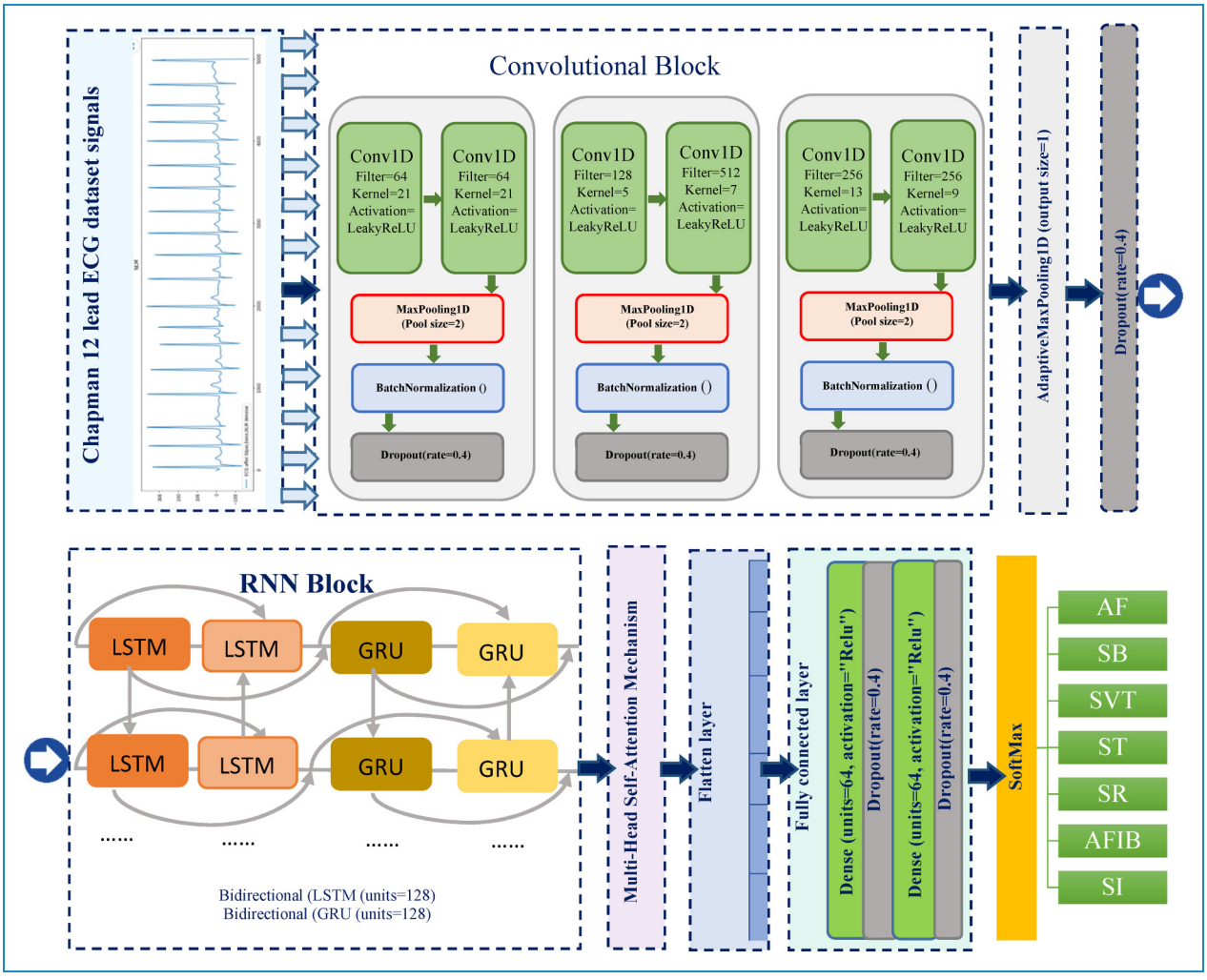


Figure 3. Schematic illustration of the proposed model to classify ECG signals.

dataset. Table 3 gives the used dataset, including seven rhythms, in detail.

To train the proposed model and check the validation, we randomly divided the dataset into three parts: 80% for training, 10% for validation and 10% for testing the model's effectiveness. Training data were balanced by the SMOTE–Tomek method. The model was trained with 200 epochs. The Adam optimizer with a learning rate of 0.001 and the categorical accuracy and F1Score evaluated the training in our proposed system.

ECG signals were classified by the Hybrid CNN-BILSTM-BiGRU Model, and then, the model was trained by using FL. We did not use the early stop criteria to compare the leads' performances during the same epochs. Subsequently, this model was evaluated only once on the test dataset. The goal was to prevent information leaks into the model due to multiple validation processes, which reduces the reliability of the model even though the model is not directly trained on the validation data.[10] Only a single efficient model was used for all the experiments. The hyperparameters of the model were not altered during training. The total parameters of the proposed model were 3,605,384; there were 3,603,720 trainable parameters and 1664 non-trainable parameters introduced by the batch-normalization layer. Table 4 shows the settings of the model parameters to obtain the best classification accuracy. We verified the effectiveness of the proposed model structure by comparing it with state-of-the-art methods.

Hardware and software

The constructed deep learning models were trained on a computer with AMD Ryzen 7 5800H with Radeon Graphics running at 3.20 GHz, an NVIDIA GeForce GTX 1650 GPU and 16 GB of memory. The DNN model was constructed using the Keras (v. 2.11.0) deep learning library and the TensorFlow (v. 2.11) framework.

Performance matrices

Our model classified seven classes of arrhythmia. To measure the overall performance of the proposed algorithm,

Table 2. The structure of the constructed CNN-BILSTM-BiGRU model.

Layer (Type)	Output Shape	Param #
Conv1D	(None, 453, 64)	1408
Conv1D	(None, 447, 64)	28,736
MaxPooling1D	(None, 223, 64)	0
Batch normalization	(None, 223, 64)	256
Dropout	(None, 223, 64	0
Conv1D	(None, 219, 128)	41,088
Conv1D	(None, 213, 512)	459,264
MaxPooling1D	(None, 106, 512)	0
Batch normalization	(None, 106, 512)	2048
Dropout	(None, 106, 512)	0
Conv1D	(None, 94, 256)	1,704,192
Conv1D	(None, 86, 256)	590,080
MaxPooling1D	(None, 43, 256)	0
Batch normalization	(None, 43, 256)	1024
Adaptive MaxPooling1D	(None, 1, 256)	0
Dropout	(None, 1, 256)	0
Bidirectional LSTM	(None, 1, 256)	394,240
Bidirectional GRU	None, 1, 256)	296,448
Attention	(None, 1, 256)	65,537
Flatten	(None, 256)	0
Dense	(None, 64)	16,448
Dropout	(None, 64)	0
Dense	(None, 64)	4160
Dropout	(None, 64)	0
Dense	(None, 7)	455
Total params: 3,605,384. Trainable params: 3,603,720. Non-trainable params: 1664.		

Table 3. The used dataset includes seven rhythms

Acronym Name	Number of Total Samples N (%)	Frequency
AF	Atrial flutter	437
SB	Sinus bradycardia	3884
SVT	Supraventricular tachycardia	582
ST	Sinus tachycardia	1563
SR	Sinus rhythm	1825
AFIB	Atrial fibrillation	1778
SI	Sinus irregularity	397
Total		**10,466**

Bold value represents the best performance.

we calculated accuracy, recall, precision and F1-score. These performance metrics are described below:

Accuracy (acc) expresses the percentage of the correctly predicted number in the total dataset and can be used to judge how often the model is correct. True positives (TP) refer to the number of correctly identified positive instances; true negatives (TN) represent the number of correctly identified negative instances and false positives (FP) and false negatives (FN) denote the number of incorrectly identified positive and negative instances, respectively. However, the Chapman–Shaoxing ECG dataset is imbalanced, so accuracy cannot be used as a good performance metric. Therefore, recall or sensitivity is utilized to measure the probability that the model correctly predicts the class as positive. In addition to recall, precision is a useful metric for evaluating the performance of classification models. It measures the frequency at which a model predicts a class as positive relative to the total number of positive instances across all classes. It provides a measure of the model's ability to make accurate positive predictions. The F1-score is a weighted average of recall and precision, offering an overall measure of a model's performance; it considers its ability to correctly identify positive instances and minimize false positives. The formulas for each performance metric are shown in Equations (1)-(4), respectively.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

Table 4. Settings of the model parameters.

Network Layers	Optimizer	Learning Rate	Dropout	Epoch	Batch Size	Cost Function
25	Adam	0.001	0.4	200	128	Focal loss

$$\mathrm{F1-score} = \frac{2_recall_precision}{recall + precision} \quad (4)$$

Model performance

Figure 4 depicts the training and validation graphs obtained from the training process.

The Chapman ECG dataset consists of a 12-lead ECG sample with a sampling frequency rate of 500 Hz for a duration of 10 s. As shown in Figure 4, the models were trained separately for each lead, resulting in a total of 12 models, each dedicated to a specific lead. The dataset distribution ensures an equal number of samples and 5000 data points per sample for each lead. This approach enables individual analysis of the unique features associated with each lead, facilitating the examination and optimization of the model's performance in detecting specific patterns and characteristics within each lead. We present the performance of each lead separately.

The developed model performed well in examining lead epoch classification. After 200 epochs of training, the categorization accuracy values for the dataset were almost 100%, with a relatively flat curve. It is evident from these graphs that the proposed model exhibited promising results for all lead ECG signals during training. Figure 5 depicts the training, validation and testing performance for arrhythmia classification, respectively. This experimental analysis was accurately calculated. Each evaluation metric is labeled with a different color in these figures, and each bar on each figure represents the accuracy of each ECG lead. The best validation and testing accuracy performance was archived by lead II (0.9845 validation accuracy and 0.9857 testing accuracy). The proposed model achieved a lower performance on lead V4 (0.9662 validation accuracy and 0.9599 testing accuracy).

After the training process, the performance of the proposed model was tested on the test sets. The trained model was applied to the test records and yielded promising results on unseen data and confusion matrices obtained during the test process. The confusion matrix plays an essential role in evaluating a model's performance. The testing-based confusion matrix for arrhythmia categorization using the dataset is displayed in Figures 6 and 7, which indicate the number of correctly predicted samples against incorrectly predicted samples. In these figures, the model achieved 0.9857 accuracy on lead II and 0.9599 accuracy on lead V4 signals.

In general, this table shows that the proposed method had an appropriate distribution regarding all classes with no emphasis on a specific class with overfitting and no bad performance concerning a specific arrhythmia class. In this matrix, most wrong states occurred between AF and AFIB arrhythmia classes, as well as SR and SI classes. As far as medical science is concerned, this miscalculation between two rhythm classes is not crucial because the diagnosis of these two arrhythmias can be reached simultaneously.

According to the confusion matrices, we calculated several performance metrics for each lead, including precision, recall and F1-score. The precision performance of each class based on each lead is presented in Table 5.

The highest precision values were obtained using lead aVF to detect SB (100%). The proposed model achieved the lowest precision performance on lead V4 (0.9084) to detect AF. The recall performance of each class based on each lead is given in Table 6. The highest recall performance values were obtained using lead II and lead aVF to detect SVT (100%). The proposed model achieved the lowest recall performance on lead V3 (0.9252) to detect AFIB. Table 7 shows the F1-score performance of each class based on each lead. The highest F1-score performance values were obtained using lead aVF to detect SVT (99.61%). The proposed model achieved the lowest recall performance on lead V4 (0.9237) to detect AF. It is interesting to know that lead II and lead aVF depicted a better performance in all of these tables. The results are given in Tables 5–7.

Discussion

This paper introduced a new Hybrid CNN-BILSTM-BiGRU algorithm that utilizes FL cross entropy and a multi-head self-attention mechanism to address the challenging problem of classifying various cardiac arrhythmias based on the Chapman–Shaoxing dataset. The SMOTE–Tomek technique was also employed to mitigate the effects of imbalanced data and detect and classify seven types of cardiac arrhythmia based on 12-lead ECG signals. Several studies have proposed models for diagnosing arrhythmias using this public dataset; thus, the advantage is that our results are comparable with those of previous studies on this database.

Numerous hybrid and non-hybrid deep learning models have been proposed in the literature; the distinction of our novel approach lies in the fact that no previous study developed a hybrid CNN-BILSTM-BiGRU algorithm by using FL cross entropy and a multi-head self-attention mechanism based on the Chapman–Shaoxing dataset.

In a comprehensive analysis of studies presented in Table 8, the performance of deep learning models for

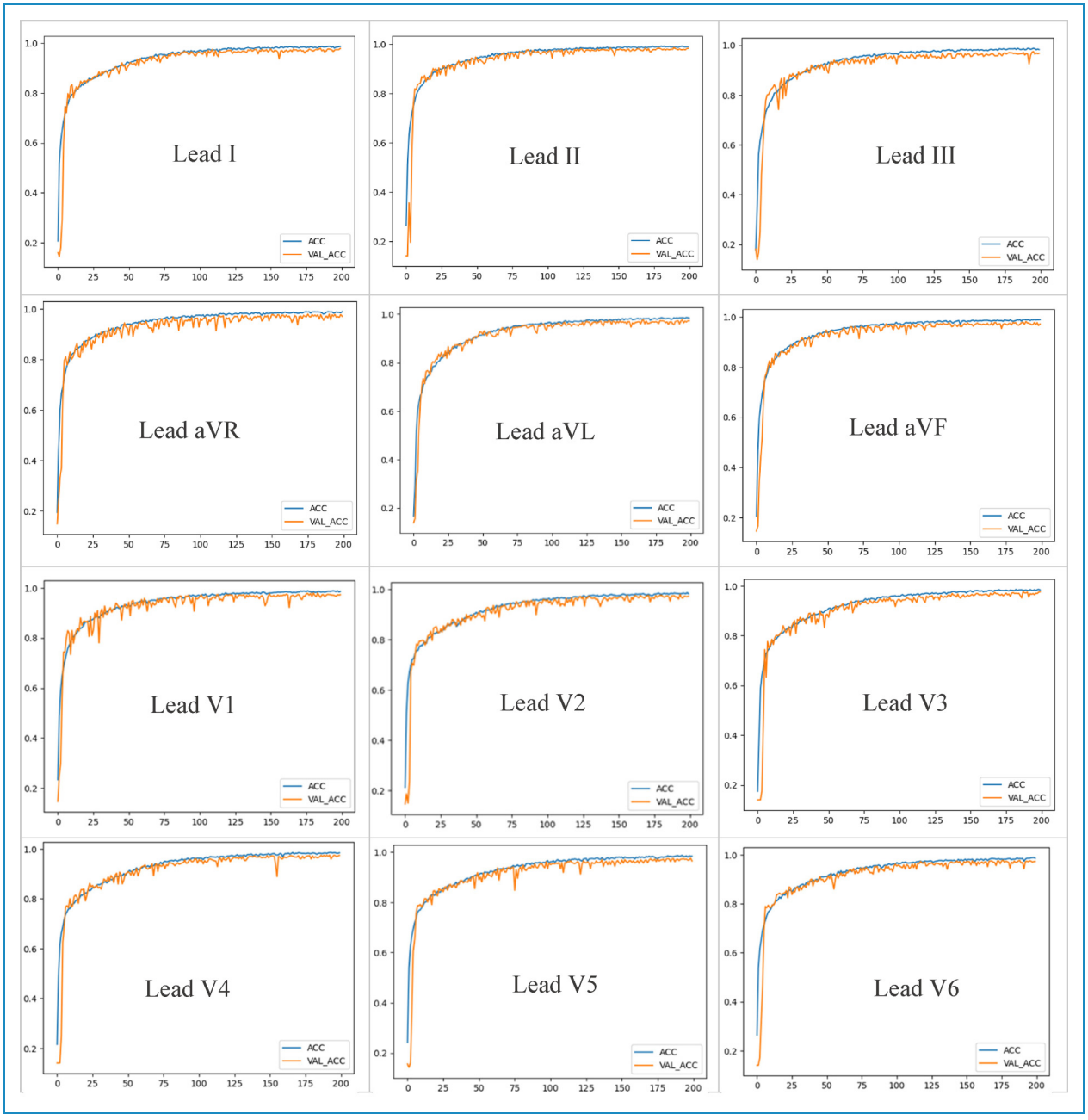


Figure 4. The training and validation graphs obtained from the training process of each lead separately.

cardiac arrhythmia detection across various databases was reported. By comparing the performance of our proposed model with the models presented in several other articles, we have demonstrated the effectiveness and competitiveness of our approach in detecting cardiac arrhythmias.

For instance, a study[5] based on the Chapman dataset used the CNN model to generate deep spatial features based on raw ECG signals of separate single leads. Then, the output of the CNN model was allocated to the LSTM model to generate the deep temporal features. This model, based on 10,436 patients, achieved a 92.24% accuracy for detecting seven classes, and based on 10,588 patients, it revealed an accuracy of 96.13% for detecting four classes. The algorithm proposed herein yielded a higher accuracy of 6.33% compared to the method proposed in[5] for detecting seven classes.

Another method is based on raw ECG signals of separate single leads. The Detrending-ResNet model was introduced to detect AFIB, atrial flutter (AFL), SVT, ST and SB arrhythmias based on RR interval signals relying on Chapman ECG samples. Its results for the classification of five rhythms manifested an accuracy of 98.55% in the evaluation section.[64] Compared to the method introduced

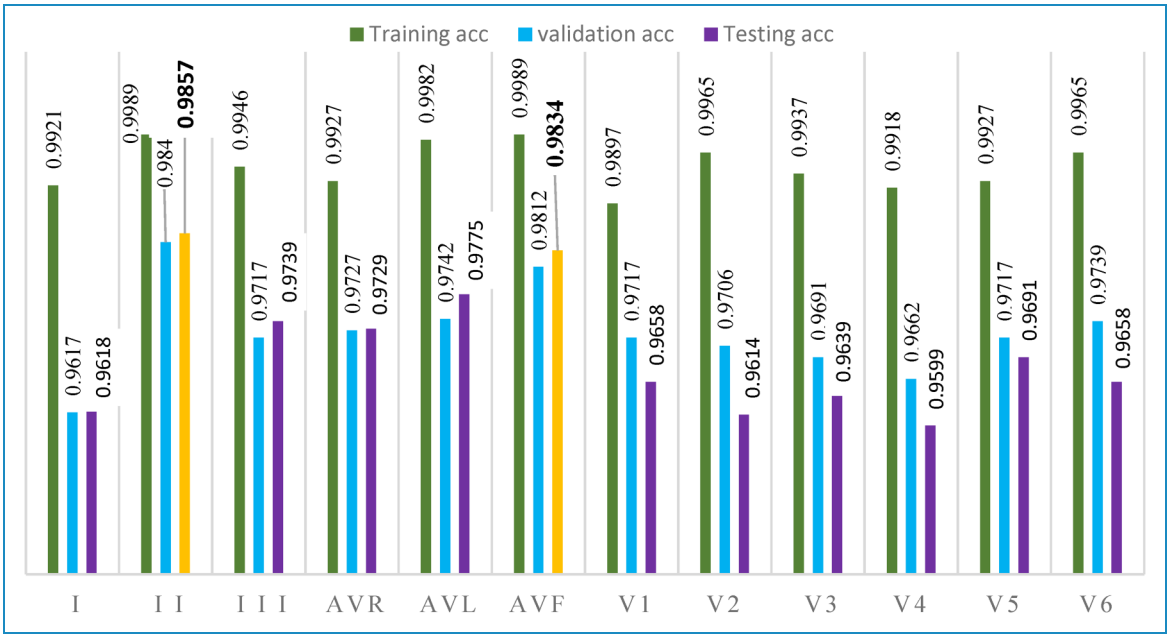


Figure 5. Training, validation and testing acc performance for all classes.

in,[64] our proposed algorithm detected one more class (SI) and yielded a higher accuracy. SR and SI classes are very similar, and, in most models, most wrong states occur between them, which reduces the model's accuracy. Evidence suggests that the proposed model based on raw ECG signals of separately single leads effectively amalgamated the beneficial features of CNN and BILSTM-BIGRU, culminating in superior accuracy, which was further enhanced by an attention mechanism.

Our model processed each lead individually and was trained with single-lead ECG records, but recently, fusion algorithms have been used in the literature to diagnose arrhythmia by fusing 12-lead data. This method has been proposed in.[65–68] In a study,[65] after fusing the 12-lead data, CNN models were coded by genetic programming. CNN trees were responsible for learning deep structural features from functional data extracted from 12 leads. According to the evaluation results, the proposed method yielded 97.60% accuracy based on the Chapman dataset. In another study, interpretable meta-structural learning was proposed in combination with CNN models encoded as the evolutionary trees of genetic programming (GP) algorithms. The performance of the proposed classification model constructed for lead III ECG had 96.92% accuracy and, for the fusing of 12-lead ECG, exhibited 98.29% accuracy for the classification of seven types of arrhythmias based on the Chapman dataset recorded from 10,646 patients. We did not combine the performance of all leads, and our proposed model was trained with single-lead ECG records but yielded higher accuracy than the methods proposed in.[65,66]

The teacher and student models were trained in another recently introduced method. The teacher model was trained with 12-lead ECG records, whereas the student model was trained with single-lead ECG records; the student's accuracy was 98.13%, and the teacher model constructed with the Chapman12-lead ECG dataset yielded a 98.96% accuracy for seven rhythm classes. Regardless of the lead selected from the 12 leads, the student model performed almost similarly to its teacher and exhibited only a small drop of 0.83% in accuracy. Our proposed model yielded a higher accuracy of 0.44% compared to the student method proposed in Sepahvand and Abdali-Mohammadi,[67] regardless of the lead selected from the 12 leads. According to the results displayed in the learning curve graph, the model converged when the number of epochs reached 150, whereas the proposed model in[67] converged when the number of epochs reached 400. In another method, the relationships of 12 ECG leads were measured using mutual information (MI) indices extracted from the ECG leads as an adjacency matrix, illustrated by the developed graph convolutional network (GCN) and included in the ECG-based diagnostic method. The GCNMI structure with 15 layers achieved 99.71% accuracy.[68]

In another study, Sadeghi et al. introduced CNNs with SE-Residual blocks and a self-attention mechanism to detect left-bundle branch blocks on the PhysioNet 2020 Challenge database with 10,344 samples from 12-ECGs. This model yielded an accuracy of 98.91%.[69] However, in the present study, we employed a combination model to detect seven types of arrhythmias.

In studies utilizing the MIT-BIH database, only data from lead II are used for the analysis and classification of cardiac rhythms. Gosia Bigam et al. developed a model that includes a CNN in combination with bilateral long-term and short-term memory to detect five types of arrhythmias. This model achieved 99.51% accuracy.[70] In other

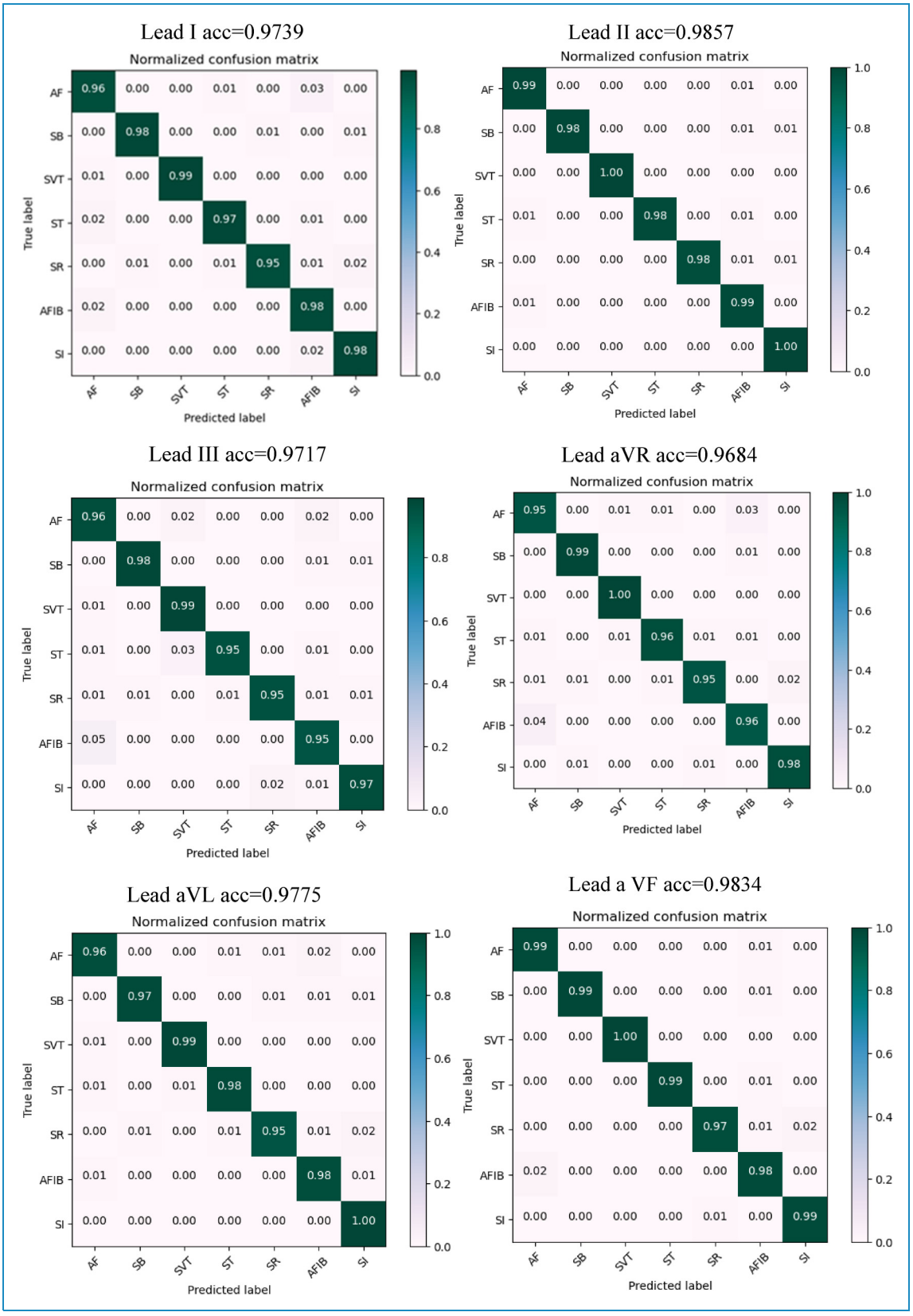


Figure 6. Testing-based confusion matrix for arrhythmia categorization using leads (I-aVF).

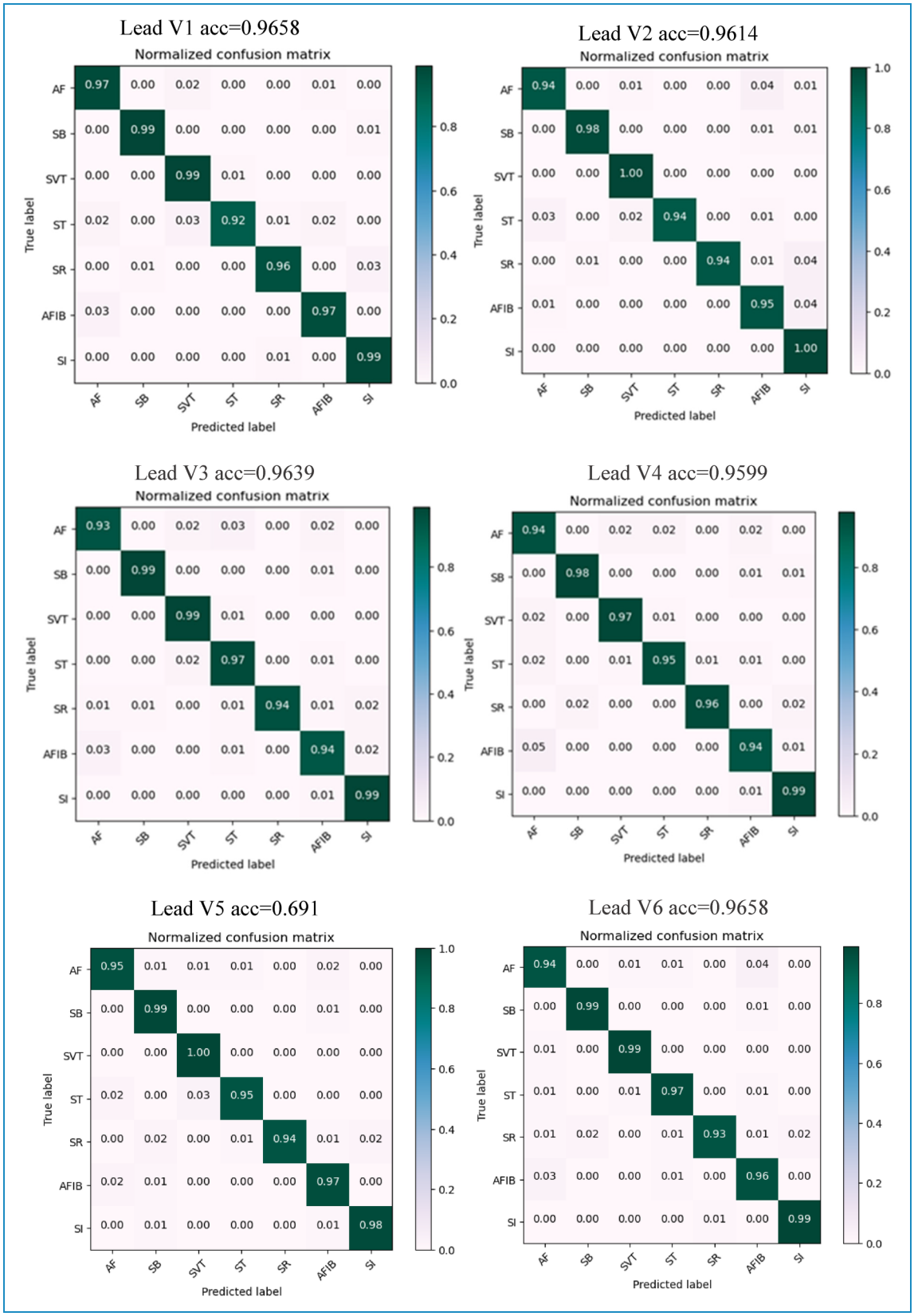


Figure 7. Testing-based confusion matrix for arrhythmia categorization using leads (V1–V6).

Table 5. The precision performance of each class based on each lead.

	AF	SB	SVT	ST	SR	AFIB	SI
Lead I	0.95.6	0.9919	0.9921	0.9843	0.9948	0.9395	0.9681
Lead II	0.9792	0.9973	0.9895	0.9974	0.9896	0.9704	0.9781
Lead III	0.9262	0.9761	0.9544	0.9213	0.9716	0.9426	0.9821
Lead aVR	0.9401	0.9738	0.9819	0.9740	0.9792	0.9547	0.9750
Lead aVL	0.9736	0.9918	0.9868	0.9793	0.9869	0.9653	0.9617
Lead aVF	0.9765	**1**	0.9921	0.9948	0.9795	0.9607	0.9826
Lead V1	0.9462	0.9867	0.9472	0.9888	0.9792	0.6577	0.9586
Lead V2	0.9496	0.9760	0.9692	0.9945	0.9947	0.9358	0.9197
Lead V3	0.9570	0.9893	0.9614	0.9398	**0.9973**	0.9513	0.9544
Lead V4	**0.9084**	0.9761	0.9634	0.9735	0.9819	0.9517	0.9658
Lead V5	0.9578	0.9512	0.9644	0.9865	0.9947	0.9556	0.9753
Lead V6	0.9375	0.9787	0.9842	0.9689	0.9841	0.9386	0.9761

Bold values represent the best performance.

studies, within the dataset, Islam et al. employed a combination of a dual-structured RNN network with a hierarchical attention mechanism based on dilated CNN to detect five types of arrhythmias. In this study, an adversarial generative network was utilized to address the data imbalance issue, resulting in an accuracy of 99.60%.[38] Another study by Islam et al. introduced the same model architecture without employing the attention mechanism, achieving an accuracy of 99.99%.[39] Besides, Khan et al. utilized CNNs to classify five types of arrhythmias. To address the data imbalance, they employed the SMOTE–Tomek technique and achieved 92.86% accuracy.[71] Rai et al. introduced a combined CNN-LSTM model to detect six types of cardiac arrhythmias based on samples from the MIT-BIH and PTB Diagnostic datasets. The data imbalance issue was addressed using the SMOTE–Tomek technique with an accuracy of 99.2%.[42]

Although our proposed model was trained with a single lead and we did not combine the performance of all leads, all lead signals were analyzed; the distinction lies in that, in the present study, the model was trained using all 12 ECG leads separately, whereas these studies solely utilized data from lead II. Although Chapman–Shaoxing is an imbalanced ECG dataset, none of the reviewed studies conducted based on Chapman–Shaoxing reported a suitable method to avoid the effects of class imbalance. For the first time, this study implemented the SMOTE–Tomek technique while also employing cross-entropy loss as a cost function to address the imbalance issue. The results demonstrated that these methods were effective in mitigating the effects of class imbalance in Chapman–Shaoxing. Furthermore, the present study utilized a multi-headed self-attention mechanism for adjusting the network structure and classifying seven arrhythmia classes. Overall, there are several justifications to improve the proposed method compared to previous methods. The accuracy of our model is high (98.57%), and it has a good generalization ability for detecting ECG arrhythmias.

Conclusion

This paper introduced a new Hybrid CNN-BILSTM-BiGRU algorithm that utilizes FL cross entropy and a multi-head self-attention mechanism to address the challenging problem of classifying various cardiac arrhythmias based on ECG data. The SMOTE–Tomek technique was also employed to mitigate the effects of imbalanced data and detect and classify cardiac arrhythmias. The experimental results suggested that the proposed classification model trained with a single lead achieved an accuracy of 98.57% for classifying seven types of arrhythmias in the Chapman ECG dataset using the lead II and samples of 10,466 patients. Furthermore, the model's performance was compared to that of state-of-the-art methods based on deep learning models, and the results demonstrated that it is highly competitive.

Table 6. The recall performance of each class based on each lead.

	AF	SB	SVT	ST	SR	AFIB	SI
Lead I	0.9584	0.9812	0.9947	0.9766	0.9571	0.9676	0.9826
Lead II	0.99	0.99	**1**	0.98	0.97	0.9801	0.9975
Lead III	0.9554	0.9813	0.9947	0.9534	0.9519	0.9403	0.9577
Lead aVR	0.9500	0.9947	0.9974	0.9715	0.9544	0.9428	0.9701
Lead aVL	0.9685	0.9759	0.9921	0.9844	0.9544	0.9701	0.9975
Lead aVF	0.9842	0.9866	**1**	0.9897	0.9696	0.9751	0.9801
Lead V1	0.9685	0.9920	0.9974	0.9145	0.9496	0.9625	0.9801
Lead V2	0.9396	0.9786	0.9974	0.9326	0.9370	0.9475	0.9975
Lead V3	0.9368	0.9893	0.9868	0.9517	0.9519	**0.9252**	0.9876
Lead V4	0.9395	0.9840	0.9735	0.9508	0.9620	0.9303	0.9801
Lead V5	0.9553	0.9893	0.9974	0.9482	0.9470	0.9675	0.9801
Lead V6	0.9424	0.9840	0.9894	0.9714	0.9369	0.9526	0.9851

Bold values represent the best performance.

Table 7. F1-score performance of each class based on each lead.

	AF	SB	SVT	ST	SR	AFIB	SI
Lead I	0.9544	0.9865	0.9934	0.9804	0.9755	0.9533	0.9753
Lead II	0.9843	0.9919	0.9947	0.9882	0.9782	0.9752	0.9877
Lead III	0.9406	0.9787	0.9742	0.9671	0.9616	0.9415	0.9698
Lead aVR	0.9450	0.9841	0.9896	0.9728	0.9667	0.9487	0.9726
Lead aVL	0.9711	0.9838	0.9894	0.9819	0.9704	0.9677	0.9793
Lead aVF	0.9803	0.9933	**0.9961**	0.9922	0.9746	0.9678	0.9814
Lead V1	0.9572	0.9893	0.9704	0.9502	0.9642	0.9601	0.9692
Lead V2	0.9446	0.9773	0.9831	0.9626	0.9650	0.9416	0.9570
Lead V3	0.9468	0.9893	0.9740	0.9554	0.9741	0.9381	0.9707
Lead V4	**0.9237**	0.9800	0.9684	0.9620	0.9719	0.9409	0.9729
Lead V5	0.9565	0.9699	0.9806	0.9670	0.9702	0.9615	0.9777
Lead V6	0.9399	0.9813	0.9868	0.9702	0.9599	0.9455	0.9778

Bold values represent the best performance.

Table 8. A comparison of the proposed model and some other state-of-the-art methods that used different leads.

Ref.	Study	Dataset	Number of ECG Records	Rhythm	Method	Performance (Accuracy %)
[64]	Faust et al.	Chapman	10,093	5	Detrending, ResNet	98.55
[5]	Yildirim et al.	Chapman	10,588 10,436	7 4	CNN-LSTM	**Lead II: 92.24** Lead II: 96.13
[65]	Meqdad et al.	Chapman	10,646	7	Meta CNN Trees	lead III: 96.92 12-lead fusion:98.29
[66]	Meqdad et al.	Chapman	10,646	7	CNN Trees	12-lead fusion:97.60
[67]	Sepahvand et al.	Chapman	10,436	7	Teacher model Student model	12-lead: 98.96 98.13
[68]	Andayeshgar et al.	Chapman	10,494	7	GCN-MI	12-lead:99.71%
[69]	Sadeghi et al.	PhysioNet 2020	10,344	1	SE-Residual blocks and a self-attention mechanism	91.98
[70]	Gosia Bigam et al.	MIT-BIH	48 samples half-hour	5	CNN-BILSTM	Lead II: 99.51
[38]	Islam et al.	MIT-BIH	48 samples half-hour	5	CNN-BILSTM-BiGRU-CCE	Lead II: 99.90
[71]	Khan	MIT-BIH	48 samples half-hour	5	CNN	Lead II: 92.86
[42]	RAI	MIT-BIH and PTB	48 samples half-hour	6	CNN-LSTM	Lead II: 99.2
The proposed model		**Chapman**	**10,466**	**7**	**CNN-BILSTM-BiGRU-Attention, FL**	**Lead II: 98.57** **Lead aVF = 98.34** **Lead III: 97.17**

Bold values represent the best performance.

We conducted an analysis of all lead signals separately to evaluate the model's effectiveness in diagnosing and classifying different types of arrhythmias. Although we did not combine all the leads, the results indicated that the proposed model's performance can be generalized to 12-lead signals. Our hybrid approach proved to be a significant and accurate method.

The future scope of the research team involves investigating the proposed model's performance on another dataset collected in Isfahan, Iran. This dataset contains 2047 ECG signals collected for 10 or more seconds with a sampling frequency rate higher than 500 Hz, labeled for 11 rhythms, including the same 7 rhythms plus 4 other rhythms, collected from 2020 to 2022. This endeavor aims to provide further insights into the model's performance on diverse datasets and enable a comprehensive comparison with existing approaches. By exploring these additional datasets, we aim to enhance the understanding of our model's effectiveness and potential in various diagnostic scenarios. Furthermore, the research team aims to develop a new deep learning method for detecting arrhythmias by combining the information of 12 leads.

Overall, the hybrid CNN-BILSTM-BiGRU algorithm, assisted by a multi-head self-attention mechanism and the SMOTE–Tomek technique, proved to be a highly effective approach for diagnosing and classifying different types of arrhythmias, representing a significant improvement over current methods. The experimental results showed that the proposed model combined the advantages of CNN and BILSTM-BIGRU and achieved better accuracy with the attention mechanism, which proves the effectiveness of this mechanism.

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